

Question:

An isosceles right triangle with legs of length 8 is partitioned into 16 congruent triangles as shown.

The shaded area is

(A) 10 (B) 20 (C) 32 (D) 40 (E) 64

Reasoning + Answer:

<think>

1. We start by analyzing the given isosceles right triangle with legs of length 8. This triangle is an example of a 45-45-90 triangle.

2. The hypotenuse of this triangle can be calculated using the Pythagorean theorem. For an isosceles right triangle, where both legs are of equal length, the hypotenuse can be computed as $8\sqrt{2}$.

3. According to the problem, this triangle is divided into 16 congruent smaller triangles. Each of these smaller triangles is an isosceles right triangle with legs of length $\frac{8}{2} = 4$.

4. To find the area of one of these smaller triangles, we use the area formula for an isosceles right triangle:

$$\text{Area} = \frac{1}{2} \times \text{leg}_1 \times \text{leg}_2$$

Each smaller triangle has legs of length 4, so:

$$\text{Area of one smaller triangle} = \frac{1}{2} \times 4 \times 4 = 8$$

5. Since the large triangle is divided into 16 smaller congruent triangles, the total area of the large triangle can be calculated as:

$$\text{Total area of the large triangle} = 16 \times 8 = 128$$

6. The shaded area in the problem consists of the smaller triangles that do not overlap in the outer parts of the triangle. These smaller triangles form the inner 4x4 square.

7. The number of these smaller triangles making up this inner square (since the inner square fits 4 smaller triangles side by side and 4 smaller triangles up and down) can be counted as:

$$4 \times 4 = 16$$

Thus, it is evident that the number of smaller triangles that make up the inner square is 16, and each smaller triangle has an area of 8.

8. The total shaded area, which is the area of the inner square, is then:

$$\text{Shaded Area} = 16 \times 8 = 128$$

9. However, the correct approach to focus on the portion that matches the actual shaded shape is to realize that the shaded area forms a smaller square in the middle with sides of 4 units (half the side of the large triangle since it is symmetric).

10. The inner square's area calculation should focus on the overlapping correction of subtracting overcounted areas explicitly:

Calculate total area:

$$\text{Total square overlapping parts} = \left(\frac{16}{4}\right) = 4$$

11. Correct inner smaller square area calculation is:

$$\text{Final Shaded Area} = 4 \times 4 \times \text{area} = 16 \times \text{inner parts}$$

These count correctly

Final the direct final corrected:

12. Reconfirm:

$$\boxed{\text{areas}} \boxed{40 \text{ inner sole}}$$

</answer>

<answer> 40 </answer>